

Hybrid randomized clinical trials incorporating external controls: assumptions and related recommendations

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<https://github.com/foucher-y/talks>
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Why augmenting randomized clinical trials (RCTs)?

- ▶ RCTs represent the reference protocol for treatment evaluations.
- ▶ For large sample size, the comparability of the arms and the power acceptable.
- ▶ But number of trials are based on small sample sizes:
 - ▶ Median at 75 patients for RCTs in intensive care units.¹
- ▶ Innovative methods aim to increase the power with external controls (Bayesian analysis, matching on propensity scores, generation of digital controls, etc.)
- ▶ A trade-off between decreasing the random bias but increasing systematic bias.
- ▶ **In this context of hybrid RCTs, the literature related to the causal assumptions is scarce.**

¹Anthon et al. (Journal of Clinical Epidemiology, 2019)

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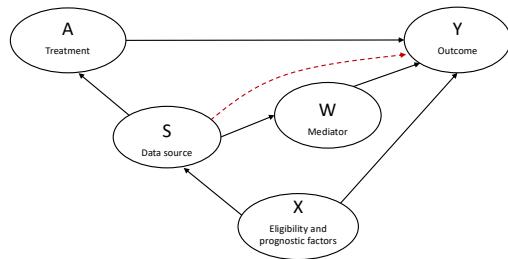
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Notations and directed acyclic graph²

- ▶ $Y = (0, 1)$: the binary outcome.
- ▶ A : the treatment arm.
 - ▶ $A = 1$ for the experimental group.
 - ▶ $A = 0$ for the control group.
- ▶ S : the data source.
 - ▶ $S = 1$ for the internal RCT.
 - ▶ $S = 0$ for the external population.
- ▶ X : the causes of both Y and S .
- ▶ W : the mediators of being included in the RCT.



²Dang et al. (arXiv, 2023)

Positivity

- ▶ Positivity violation occurs when it is certain that patients are allocated to a group.
- ▶ We need to respect in the targeted population (i.e., $\Pr(S = s|X) > 0$):

$$0 < \Pr(A = a|X, S = s) < 1 \text{ for all possible } s$$

- ▶ Because of the randomization, this condition holds for individuals in the RCT.

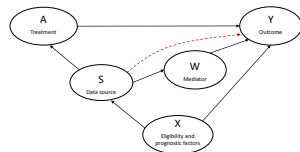
Assumption 1

Positivity holds if the individuals of the external data source have a non-zero probability to be included in the RCT:

$$\Pr(S = s|X) > 0 \text{ for all possible } s$$

Absence of unmeasured eligibility and prognostic factors

- ▶ Random assignment of A ensures that there is no confounder within the RCT (only near-confounders, i.e. chance imbalanced prognostic factors).
- ▶ However, the factors X may represent a source of bias.
- ▶ For instance, elderly patients are often under-represented in a RCT. If Y is related to mortality, this imbalance may lead to an overestimated effect of the experimental treatment.



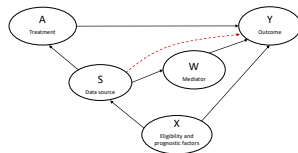
Assumption 2

The eligibility criteria and prognostic factors X are measured.

No direct effect on the outcome of being in the trial

- ▶ The average outcome for controls given factors X is the same in the RCT and external population.

$$E(Y|X, A = 0, S = 1) = E(Y|X, A = 0, S = 0)$$



- ▶ It can be violated for different schedules of visits in a trial.
- ▶ The follow-up is an example of a mediator W .
- ▶ The measurement of W can further help in correcting the related bias by modeling

Assumption 3

There is no direct effect of S on Y conditional on (X, W) .

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Historic recommendations for acceptability of (full) external controls³

The external individuals...

- ▶ ...must have been part of a recent study with the same inclusion criteria ($\sim A1$).
- ▶ ...must have been included in the same organization by same investigators ($\sim A1$).
- ▶ ...must have comparable characteristics ($\sim A2$).
- ▶ ...must be no other indications leading one to expect differing outcomes ($\sim A2$).
- ▶ ...must have the same outcomes ($\sim A3$).
- ▶ ...must have received the same treatment ($\sim A3$).

³Pocock (Journal of Chronic Diseases, 1976)

Twelve 12 recommendations according to the 3 assumptions

Assumption 1 Positivity	1.1. Inclusion and exclusion criteria of the external and internal subjects are similar. 1.2. External controls come from a concomitant or recent study. 1.3. The external study consists for a large part of the same or similar centers.
Assumption 2 No unmeasured eligibility and prognostic factors X	2.1. Factors X are available in both the internal and external data. 2.2. Factors X are measured similarly across the internal and external studies. 2.3. Factors X should include social and lifestyle dimensions.
Assumption 3 No direct effect on the outcome	3.1. Baseline of the controls is identical in the RCT and the external study. 3.2. External controls receive the same treatment as in the control ones. 3.3. Patients in the RCT have a follow-up as close as possible as the routine care. 3.4. Outcome Y are measured similarly in the RCT and the external study. 3.5. Mediators W are measured similarly in the RCT and the external study. 3.6. Missing data on the variables (X , W , Y) is rare and (completely) at random.

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Treatment regression for evaluating the positivity

- ▶ Description to check the overlap of the characteristics X between the internal and external controls.
- ▶ Modelling S according to X to identify issues when predictions are lower than ϵ or higher than $1 - \epsilon$.
 - ▶ Logistic regression.
 - ▶ Regression trees, such as the Port algorithm.⁴
- ▶ **In case of violation, the investigators may restrict the targeted population to recover positivity.**

⁴Danelian et al. (Journal of Causal Inference, 2023)

Outcome regression

- ▶ The respect of the assumptions 2 and 3 result in similar predicted outcomes for controls regardless if they were enrolled in the RCT or in the external study:

$$\Pr(Y = 1|X, S = 0, A = 0) = \Pr(Y = 1|X, S = 1, A = 0)$$

- ▶ Several methods can be used:
 - ▶ Among all the controls, modelling $\Pr(Y = 1|X, S)$ for testing residual effect of S conditionally on X .
 - ▶ Among the controls of the external source, one can model $\Pr(Y = 1|X)$ and study its calibration by using the internal controls.
- ▶ In case of violation, one can further explore if the direct effect can be explained by W :

$$\Pr(Y = 1|X, W, S = 0, A = 0) = \Pr(Y = 1|X, W, S = 1, A = 0)$$

- ▶ If not, the treatment effect may be biased.

Required sample size for such a calibration

- ▶ In our context with a binary outcome, one can propose checking that $O/E = 1$:
 - ▶ $O = \sum_i Y_i$ is the number of observed events among the internal controls, and
 - ▶ $E = \sum_i \hat{\text{Pr}}(Y_i = 1 | X = x_i, S_i = 1, A_i = 0)$ is the expected number thanks to the outcome regression.
- ▶ The required sample size for such a calibration can be estimated by:⁵

$$N_{0c} = (1 - \pi_0) / (\pi_0 SE_{\alpha/2, w}(\log(O/E))^2)$$

- ▶ π_0 is the expected proportion of events ($Y = 1$) among controls, and
- ▶ $SE_{\alpha/2, l}(\log(O/E))$ is the standard deviation associated with the width l of the targeted $(1-\alpha)\%$ confidence interval (CI) of $\log(O/E)$

⁵Riley et al. (Statistics in Medicine, 2021)

Required sample size for a superiority trial

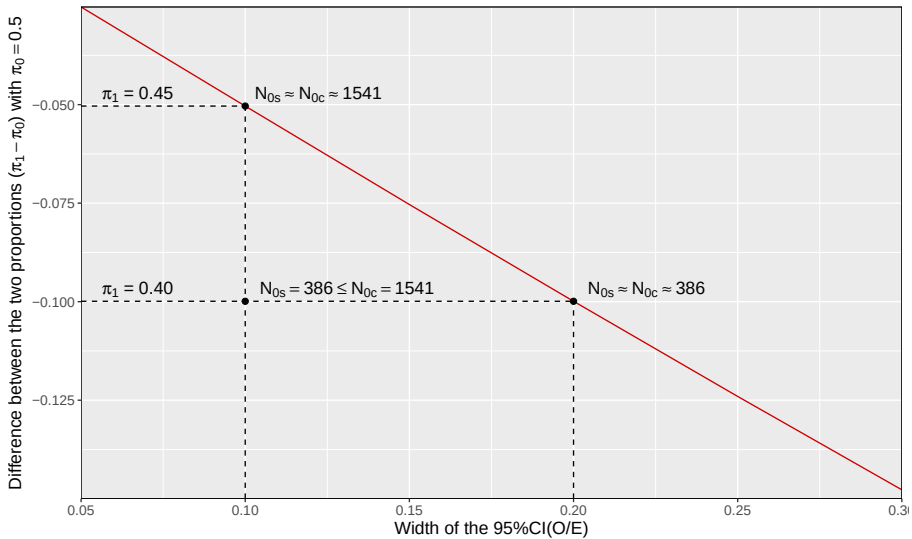
- ▶ For a usual 1:1 superiority RCT, the required sample sizes in each arm can be estimated by:

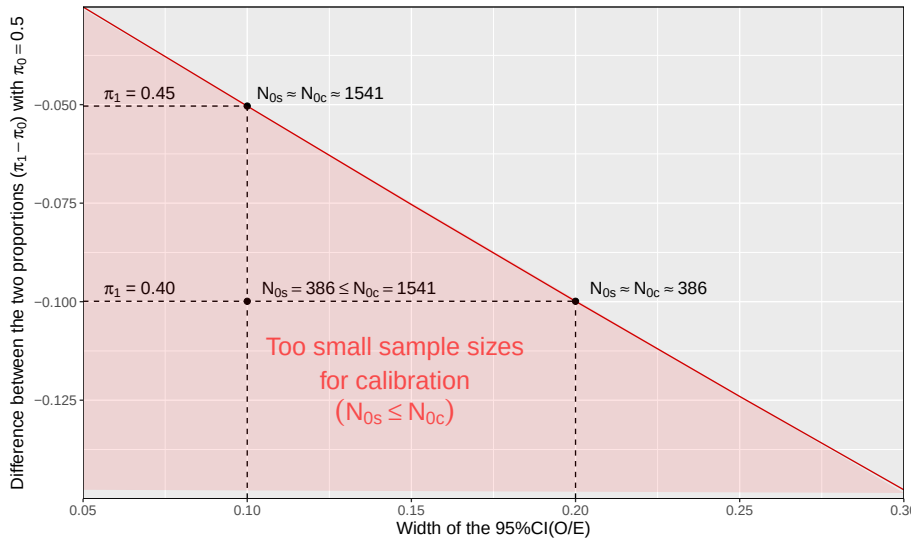
$$N_{0s} = N_{1s} = (\pi_1 - \pi_0)^2 (\pi_0(1 - \pi_0) + \pi_1(1 - \pi_1)) / (z_{\alpha/2} + z_{\beta})^2$$

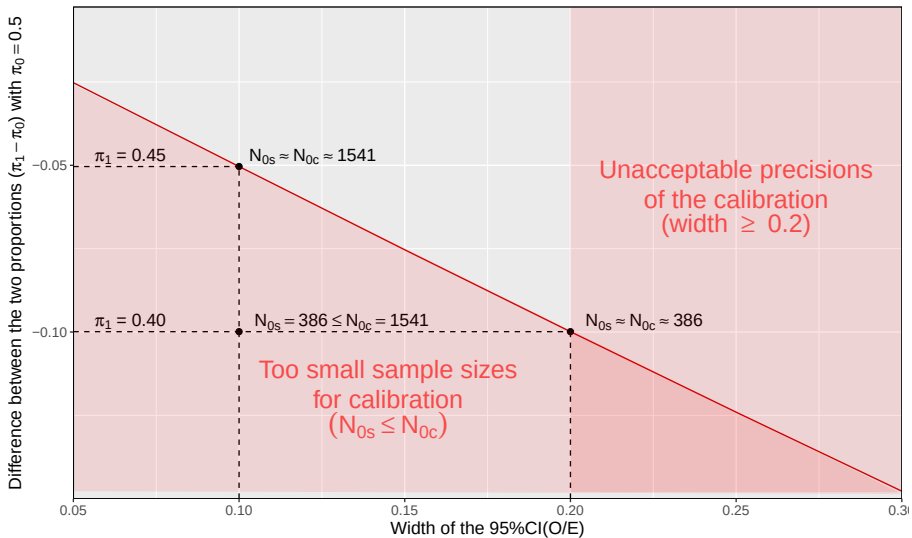
- ▶ π_1 is the expected proportion of events ($Y = 1$) among the experimental arm,
- ▶ Type-I error equals α , and
- ▶ Type-II error equals β .

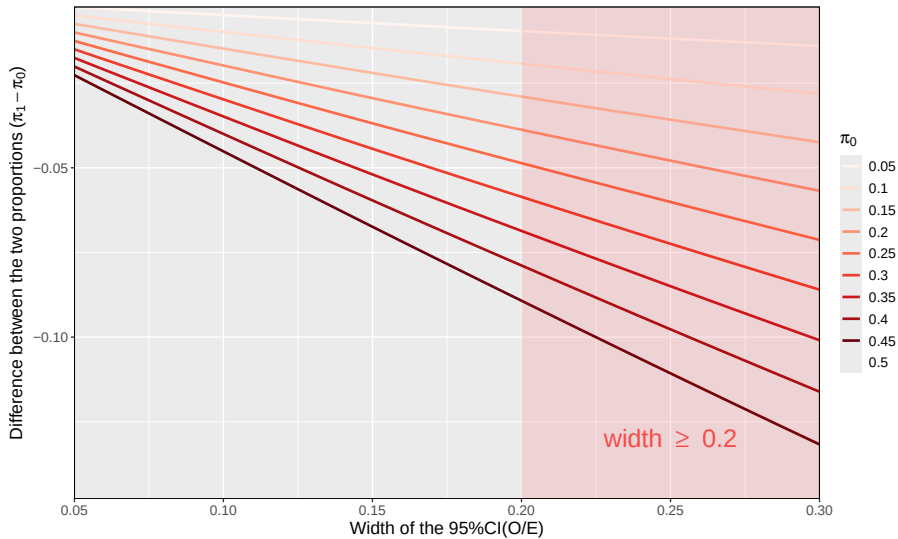
Two illustrative examples with $(\alpha, \beta) = (0.05, 0.20)$

- ▶ Example 1:
 - ▶ $(\pi_0, \pi_1) = (0.50, 0.45)$
 - ▶ Width of the 95% $CI(\log(O/E))$ equals to 0.20
 - ▶ $SE_{0.025, 0.20}(\log(O/E)) \approx 0.051$
 - ▶ Sample size of controls for the usual 1:1 superiority trial: $N_{0s} \approx 1562$ patients.
 - ▶ Sample size of controls for calibration: $N_{0c} \approx 386$ patients.
 - ▶ **The effective of internal controls will allow for investing the calibration.**
- ▶ Example 2:
 - ▶ Same data but an higher effect: $(\pi_0, \pi_1) = (0.50, 0.35)$
 - ▶ Sample size of controls for the usual 1:1 superiority trial: $N_{0s} \approx 170$ patients.
 - ▶ **The effective of internal controls will not allow for investing the calibration.**
- ▶ Example 3:
 - ▶ $(\pi_0, \pi_1) = (0.50, 0.40)$: $N_{0s} \approx N_{0c} \approx 386$ patients.









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Summary of the key points for such an hybrid trial

- ▶ External controls represent an opportunity but 3 assumptions should be respected.
- ▶ We proposed a bundle of 12 recommendations which can guide the design.
- ▶ It underlined the importance of prospectively designing both the internal and external sources as close as possible:
 - ▶ Retrospective registries or medico-administrative databases as external sources should be used with caution.
 - ▶ The prospective collection of the internal data must be designed according to the methods used for external controls.
- ▶ Paradox: While external controls may be used when the internal sample size is small, exploring the respect of the assumptions necessitates large sample size.

References

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5. Riley et al. (2021). Minimum sample size for external validation of a clinical prediction model with a binary outcome. *Statistics in Medicine*, 40(19), 4230–4251.